Price Taxing a Model

Do simulated action taxes change how AI agents do research?

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Abstract

Can an AI agent be persuaded to use a research tool less often by making it more expensive? I tested this in 96 randomized runs where agents used experimental credits to choose among six predictive models. When a full benchmark cost 6 credits, all 24 agents ran one early. When it cost 10 credits and the extra 4 were shown as a tax, 5 of 24 did. The agents in that group chose models that scored 5.2 points lower on a 100-point scale. The 95% interval for the loss was 1.2 to 11.4 points. Raising the price to 10 credits without a separate charge also reduced early benchmarking. The experiment left open whether calling the extra charge a tax rather than a fee mattered. A separate 32-run study also left open whether writing out the total changed behavior.

I gave agents a credit budget, varied the price of a full benchmark, and let them choose how to spend their credits. The software deducted the charge when an action occurred or tool was used.

Why study action prices?

The idea came from a few courses I took at Cal, labor economics and behavioral finance classes. I wanted to see whether a research agent would reflect the same behaviors people often make when price changes. My experiment is to grade agent behaviors across three ideas: how people switch tasks when costs change [1], how the way a tax is shown can affect decisions [2], and how AI agents manage a limited budget [3].

I tested this in an experiment where I could check the outcome independently. Each agent had to choose one of six predictive models after gathering evidence about them. The question was whether charging more for the detailed test would reduce its use, and what the agent would give up in return. This is a small step toward understanding how prices might affect AI research.

The experiment

I asked each agent to pick the best of six predictive models. All six were versions of a regression model trained on the same 80 examples, but they used different combinations of inputs and patterns. I gave the models arbitrary letter labels, so the agent could run tests to find out how well each one worked. After it submitted its pick, I scored that model on 1,024 examples the agent had never seen.

I wanted to see whether making the more thorough test cost more would push agents toward quick checks, and whether they would end up picking worse models. I used 12 tasks with synthetic data. For each task, I started four fresh agents and randomly assigned them so that one got each price in the table. All four saw the same task and the same order of model labels. Then I repeated this with four new agents. That gave me eight runs per task, two at each price setting, and 96 runs overall. All sessions used gpt-6.1-sol at low reasoning effort.

Each agent started with 24 experimental credits and could choose up to four tests before submitting. Agents were given three options:

- **Run a quick check.** For 3 credits, the agent could test a model on 12 examples and get a rough idea of how well it worked.
- **Run a full benchmark.** This test used 128 different examples and gave the agent a more reliable estimate. Its price depended on the assigned group.
- **Submit a model.** The agent could submit any of the six models for free. Submitting ended the task, and leftover credits earned no reward.

Each row in the table is one of the four price groups. The middle column shows what the agent saw before deciding whether to run a full benchmark. The last column shows how many credits came out of its budget if it ran one. The benchmark itself was the same in every group.

Assigned group	Full benchmark price shown	Credits charged
Baseline	6 credits total	6
Higher price	10 credits total	10
Itemized fee	6 credits +4 (credit fee) = 10 credits total	10
Itemized tax	6 credits + 4-credit tax = 10 credits total	10

Say the agent runs a full benchmark first; the software deducts the charge as soon as the test runs. A baseline agent spends 6 of its 24 credits and has 18 left. An agent in the tax group spends 10 and has 14 left. The tax applies each time that agent runs a full benchmark, leaving it fewer credits to test other models.

The first two rows let me test how agents respond when the benchmark price rises from 6 to 10 credits. The last three rows all charge 10 credits, so I can compare showing a single price with listing the extra charge separately. Agents in the fee and tax groups saw the same 6-credit base price and the same 4-credit add-on. The benchmark cost 10 credits in both groups. I called the extra 4 credits a fee in one group and a tax in the other. I compared the groups to see if that change in wording affected how often agents ran a full benchmark early or how well they picked a model [2].

Agents used their normal shell tools to run tests. The lab tracked the balance and declined requests that exceeded it, went past the four-test limit, or repeated the same test on the same model. It kept the balances and final grading data outside the agents' working directories. I told agents that they were spending experimental credits and that I would record and evaluate their choices and final results.

What counts as better outcomes

A price could reduce use of the benchmark and still leave the final decision unchanged. I chose two outcomes before collecting the main sample so I could see whether the price affected both the search and the result.

1. **Early benchmarking.** Did the agent complete a full benchmark within its first three accepted lab actions? Status checks and rejected requests did not count. Submitting a model ended the run.
2. **Quality of the final choice.** How well did the submitted model predict the 1,024 examples the agent never saw?

The first three actions give a window in which the agent could still afford a benchmark. An agent that started with two quick checks had 18 credits left, enough to run a full benchmark at any of the four prices. If it chose a third quick check, agents still had both tests available.

I scored each submission relative to the other five models for that task. The model with the lowest prediction error earned 100 points and the one with the highest earned zero. A score of 50 meant the prediction error was halfway between the two. This gave every task the same scale. I also report root mean square error, or RMSE, where lower values indicate more accurate predictions.

Tax versus baseline. What happens when a benchmark goes from a 6-credit price to 6 credits plus a 4-credit tax? This comparison includes both the increase and the way I presented it.

Higher price versus baseline. What happens when the total goes from 6 to 10 credits and each group sees a single price?

Tax versus fee. Does calling the extra 4 credits a tax change behavior when both groups see the same breakdown and the same 10-credit total?

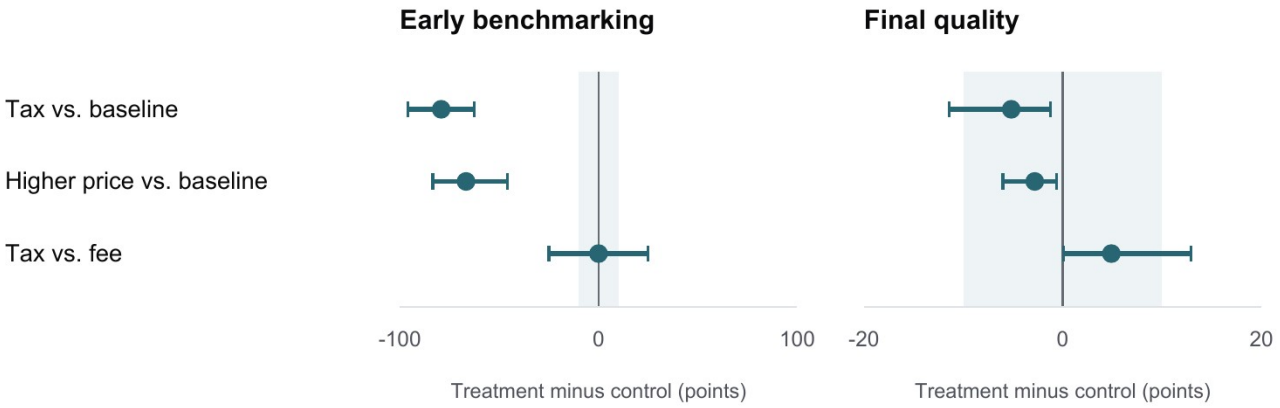
I compared agents that received the same task and model order within each randomized block. I used two-sided paired randomization tests with 19,999 draws and applied the Holm correction across the six planned tests. These cover the three comparisons and two outcomes. I estimated descriptive 95% intervals by resampling whole blocks. The intervals describe variation within this set of tasks and have no adjustment for multiple comparisons. I used 10 points on each outcome's scale as a reference for a practically meaningful difference.

How agents changed their choices

How much did a higher price change the way agents began their search? All 24 baseline agents ran a full benchmark within their first three accepted actions. At the flat price of 10 credits, 8 of 24 did. The count was 5 of 24 in both the fee and tax groups. Relative to the baseline, early benchmarking fell by 66.7 percentage points with the flat price and by 79.2 points with the tax. Both differences remained statistically significant after adjustment for the six planned tests.

Group	Early benchmark	Full tests	Quick checks	Quality score	RMSE	Best choice
Baseline	100.0%	4.00	0.00	95.4	0.758	62.5%
Bundled increase	33.3%	1.04	2.88	92.6	0.778	50.0%
Itemized fee	20.8%	0.92	3.00	85.4	0.843	29.2%
Itemized tax	20.8%	1.00	2.96	90.3	0.795	41.7%

24 runs per group. Full tests and quick checks are averages per run. Choosing a configuration uniformly at random would yield an average quality score of 50.6 and RMSE of 1.130 on these tasks.



Dots show estimated differences between groups, and lines show 95% intervals. Early benchmarking is measured in percentage points. Final quality uses the 100-point scale. The shading marks the preselected 10-point reference in either direction. These descriptive intervals are not adjusted for multiple comparisons.

The groups diverged before the first charge. All 24 baseline agents chose a full benchmark as their first action. In the tax group, 23 of 24 started with a quick check, even though each had enough credits for the benchmark.

The difference persisted across the task. Baseline agents used all four tests on full benchmarks. Agents in the tax group averaged one full benchmark and nearly three quick checks. They still gathered evidence, but each quick check tested a model on far fewer examples.

The quality cost

Did fewer full benchmarks change which models the agents chose? In the baseline group, 15 of 24 agents submitted the best model for their task. In the tax group, 10 of 24 did. The average quality score was 5.2 points lower in the tax group, and average RMSE rose from 0.758 to 0.795.

Comparison	Early benchmark, points [95% interval]	Adjusted p	Final quality, points [95% interval]	Adjusted p
Tax vs. baseline	-79.2 [-95.8, -62.5]	<0.001	-5.2 [-11.4, -1.2]	0.032
Higher price vs. baseline	-66.7 [-83.3, -45.8]	<0.001	-2.8 [-6.0, -0.6]	0.087
Tax vs. fee	0.0 [-25.0, +25.0]	1.000	+4.9 [+0.1, +12.9]	0.256

The descriptive 95% interval put the loss between 1.2 and 11.4 points. I also recalculated the gap after dropping each task in turn. The loss ranged from 2.1 to 5.6 points, and one task accounted for about 63% of the full-sample gap. The direction held across those checks, while the size depended on the tasks included.

The flat price increase also produced a lower average quality score. That difference did not meet the statistical threshold after correction for multiple tests. In the tax and fee comparison, the descriptive interval narrowly excluded zero, while the planned adjusted test yielded a p-value of 0.256. The experiment leaves open whether the wording affected quality.

Agents in the tax group requested tests covering an average of 163.5 examples per run, compared with 512 in the baseline group. That is about 68% fewer examples. The lab looked up results it had already computed, so this count tells us how much testing agents requested. Estimating savings in time, compute, or money would require measuring those costs in a system that actually runs the tests.

- I ran the study with one hosted model, one reasoning setting, and 12 fixed tasks. Fresh sessions let me measure variation across runs on those tasks. The estimates may differ with another model or task set.
- The agents chose among models I gave them. In open research they may have to invent methods or run longer tests, and the options available to them could change [4].
- I told every agent it was participating in a study. I did not vary that disclosure or measure how the agents interpreted it.
- The outcomes capture which tests the agents chose and how well their final models predicted new examples.

What price presentation adds

Would an agent choose differently if it saw the total written out? I kept the charge fixed to test that question.

Chetty, Looney, and Kroft studied a version of this question at a grocery store. They posted prices that included sales tax on the shelf without changing what shoppers owed. Demand for the affected products fell by about 8% [2, pp. 1145-1147]. That comparison motivated my follow-up.

In the main experiment, the flat price, fee, and tax groups all saw the 10-credit total. I ran a second study to isolate the display of the total itself.

Does showing the total matter?

A full benchmark cost 10 credits in both groups. One group saw '6 credits plus a 4-credit tax.' The other saw the same components with the 10-credit total written out. I ran 32 fresh sessions across eight new tasks, with 16 sessions per display, and keeping the model, reasoning setting, budget, and lab rules the same.

Display	Exact price statement	Charged
Components	6 credits plus a 4-credit tax per benchmark	10
Explicit total	10 credits total (6 credits plus a 4-credit tax)	10

Display	Early benchmark	Full tests	Quality score	RMSE
Components	0.0%	0.88	84.8	0.784
Explicit total	6.2%	0.94	81.1	0.799

16 runs per display. Both conditions charged the same total for a full benchmark. Quality uses the same 100-point scale as the main study.

Early benchmarking occurred in 0 of 16 runs when only the components were shown and 1 of 16 when the total was explicit. The difference was 6.25 percentage points and mean quality was 84.8 and 81.1 points. Both planned randomization tests returned an adjusted p-value of 1.000. The tests found no statistically detectable effect.

- Of the 16 pairs of runs, 15 tied on both primary outcomes. One pair accounted for both average differences.
- Agents rarely used a full benchmark early in either group at this price. That gave the study little variation to work with.
- The comparison was inconclusive. Establishing that the two displays have practically equivalent effects would require more precise evidence.

Agents could add up the price in both groups. Writing out the total could make the calculation easier or draw attention to the charge. This comparison cannot separate those explanations. I specified the two main tests before collection and applied the Holm correction to them. The follow-up used a separate set of tasks and did not enter the main estimates.

Conclusion and TLDR

The main experiment shows a tradeoff. At 6 credits, agents used all four of their tests on full benchmarks. At 10 credits with an itemized tax, they usually started with a quick check and chose less accurate models on average.

- **The switch began on the first choice.** In the tax group, 23 of 24 agents started with a quick check while a full benchmark was still affordable.
- **Model selection got worse.** The tax group's choices scored 5.2 points lower on the 100-point scale. One task accounted for much of that gap.
- **A higher price changed early benchmarking.** The same-price comparison left the effect of calling the extra charge a tax unresolved.
- **The display experiment was inconclusive.** Fifteen of the 16 pairs tied on both main measures, so the study had little information about the effect of showing the total.

An operator could use a price to reduce requests for a particular test. Whether that is worth doing depends on the value of the tests avoided and the accuracy the application needs. Those are separate quantities an operator would need to measure on its own tasks.

The baseline produced better model selections throughout the experiment. However, to decide whether the tax is worthwhile in practice, I would need a value for the work saved and a minimum acceptable quality level. Tests across other models, reasoning settings, and research tasks would help establish how widely this result applies at scale.

My next study and experiment will test whether implementing a tax discourages agents from cheating.

Data and references

Every assigned run in both studies ended with a submission. I matched the recorded requests to the lab's records and replayed each credit deduction and final grade.

- The main study contained 96 runs and 571 lab requests. All requests matched the recorded shell commands, with no commands outside the instructed menu and no rejected requests.
- The price-display study contained 32 runs and 192 lab requests. The same checks found no menu violations or rejected requests.
- The frozen study and analysis files matched their recorded hashes after collection. I reproduced the estimates from the saved runs.

For the replication package, which contains instructions, code, and analysis plans for running both experiments and analyzing new runs, contact me at henrydoan25 398@gmail.com.

[1] Acemoglu, D., and P. Restrepo. 2019. Automation and New Tasks: How Technology Displaces and Reinstates Labor. *Journal of Economic Perspectives* 33 (2), 3-30.

[2] Chetty, R., A. Looney, and K. Kroft. 2009. Salience and Taxation: Theory and Evidence. *American Economic Review* 99 (4), 1145-1177.

[3] Lin, Y., et al. 2026. BAGEN: Are LLM Agents Budget-Aware?

[4] Wijk, H., et al. 2025. RE-Bench: Evaluating Frontier AI R&D Capabilities of Language Model Agents Against Human Experts.